

(EST MINI PROJECT)

AI FOR A GREENER FUTURE

AI Project Report

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ABSTRACT

Waste segregation is essential for effective recycling and reducing environmental pollution. However, manual sorting can be time-consuming and prone to errors. The objective of this project is to develop an image-based waste classification model using a Convolutional Neural Network (CNN) to identify different types of waste automatically.

The model is developed using TensorFlow and Keras. The dataset is divided into 80% training data and 20% validation data. Images are resized to 128 × 128 pixels and normalized before being processed by the CNN, which learns image features and predicts the appropriate waste category.

The model is trained for 10 epochs using the Adam optimizer. Its performance is evaluated using accuracy, loss graphs, a confusion matrix, and a classification report. The trained model is also tested on sample images and saved for future use.

The significance of this project lies in its potential to make waste segregation faster, more consistent, and less dependent on manual effort. It can support better recycling practices, reduce the amount of waste sent to landfills, and contribute to sustainable waste management. Since the notebook has not yet been executed, the final accuracy and classification results will be added after training. Future improvements could include testing real-world images and integrating the model with an automated waste-sorting system.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Dr. Jyoti , for her valuable guidance, encouragement and support throughout our project, HDFC Bank Loan Approval Prediction Using Logistic Regression.

Her clear explanation of the machine learning workflow, from defining the problem and preparing the data to evaluating and interpreting a model, helped us understand that a project is more than writing code. Her emphasis on justifying every choice, from preprocessing to algorithm and metrics, shaped how we approached each stage. Her feedback and suggestions helped us improve our analysis and how we present it.

We are also thankful to Thapar Institute of Engineering & Technology for providing us with the opportunity and resources to carry out this project. We thank the Kaggle community for making the Loan Prediction dataset publicly available, and the developers of the open-source Python libraries that made this work possible. Finally, we thank our families and friends for their constant support and motivation.

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1.1 PROBLEM

The problem actually begins where people put waste in the wrong bin without even knowing how harmful it can be for the living beings. All types of wastes, recyclables, organics, etc eventually gets mixed which contaminates recycling and sends hazardous e-waste to landfill.

1.2 OBJECTIVE

The aim is to build and train a CNN that predicts E, O or R from a 128 × 128 image. It is evaluated with accuracy, per-class precision, recall and F1, and a confusion matrix, not one number. The work also tests how stable the result is across training runs, how it does on a held-out test set, and how it compares with simple baselines and design variants. Finally, it demonstrates a prototype that returns a category, a confidence and sorting advice.

1.3 OVERVIEW

The pipeline loads the waste images, resizes them, scales the pixels to 0–1 and splits them for training and evaluation. A CNN is trained on them. In the notebook it reached 95.29% on 85 validation images, but repeated runs gave 75–87% and the held-out test gave about 76–80%. The CNN also did not clearly beat a random forest on colour features (81.5%). The report therefore presents the headline figure alongside these stricter results and recommends transfer learning and better data as next steps.

DATASET

2.1 SOURCE

The project uses the publicly available Waste Segregation dataset from Kaggle, stored in the file waste_dataset_small.zip. It contains 2,549 JPEG photographs of waste items, arranged in folders by split and class. Each image belongs to one of three classes, E, O or R, and the name of the folder it sits in is its label. The dataset is already divided into a TRAIN folder and a TEST folder.

2.2 DATASET SIZE

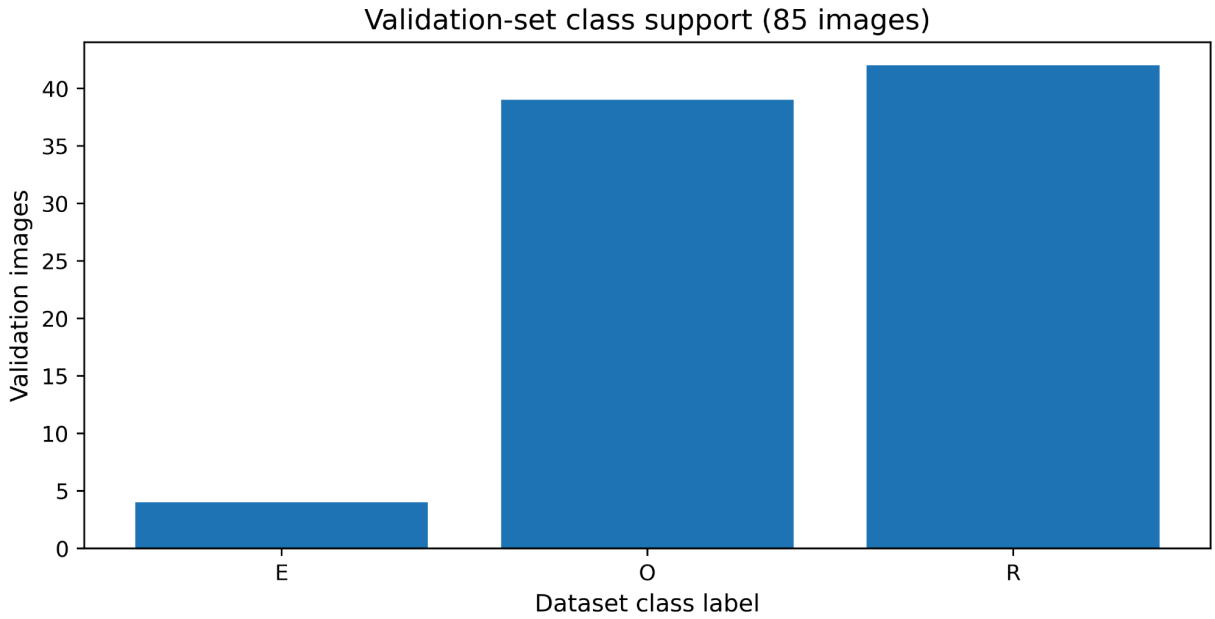
The full dataset has 2,549 images: 2,123 in the TRAIN folder and 426 in the TEST folder.Each image is resized to 128 × 128 pixels with 3 colour channels, so every image gives the model 49,152 input values.

2.3 INITIAL INSPECTION

Before building anything, we'd open the ZIP and look at how the data is laid out. There are two folders, TRAIN and TEST, each with E, O and R sub-folders, 2,549 photos in all. We'd count the images per class and notice straight away that E is tiny (149) next to O and R (1,200 each), then flip through a few random samples to confirm what E, O and R mean. We'd also check the basics: that every file opens, how big the pictures are (E images turn out to be about twice as wide as the others), and whether any duplicates slip between TRAIN and TEST (three do). That tells us what we're working with before we resize, scale and train.

2.4 DATASET CHARACTERISTICS

In the TRAIN folder, E and O each have 1,000 images, while R has 123. In the TEST folder, E has 26 images and O and R each have 200. Such differences can affect the metrics and make overall accuracy less informative for the smaller classes.



2.5 LITERATURE

2.5.1 CNN

CNN is a deep learning model used to identify objects in images. It learns visual features such as shapes, edges, and textures, helping the system recognise different types of waste automatically.

2.5.2 IMAGE PROCESSING

Image classification is the process of assigning an image to a particular category. In this project, the model analyses waste images and predicts the category that best matches the image based on patterns learned during training.

2.5.3 POOLING LAYERS

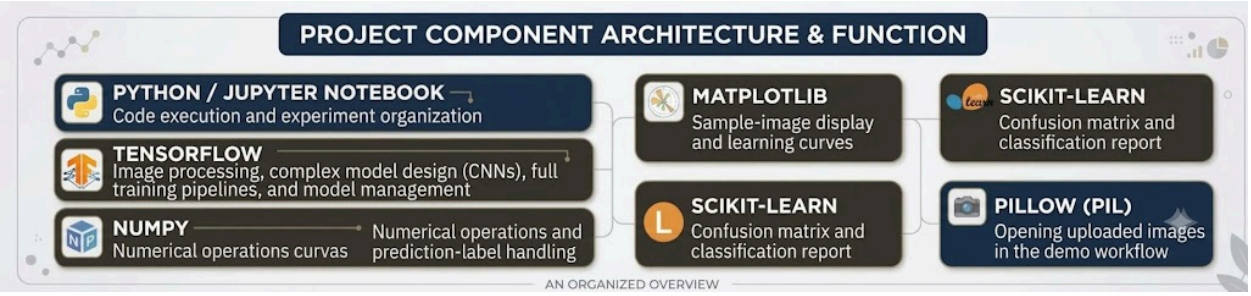
Pooling layers reduce the size of feature maps while keeping the most useful information. This helps the model process images more efficiently and reduces the amount of computation needed during training

2.5.4 DENSE LAYERS

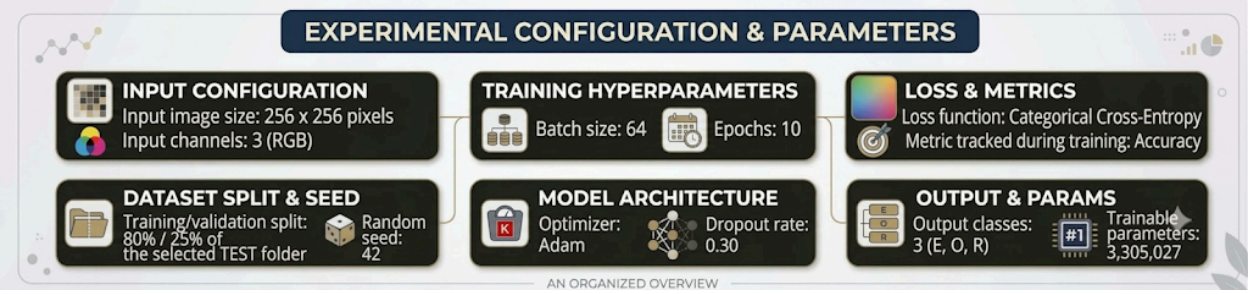
Dense layers combine the features extracted by earlier layers to make the final prediction. The output layer uses Softmax to produce scores for each waste category, and the category with the highest score is selected as the model's prediction.

METHODOLOGY

3.1 SOFTWARE AND LIBRARIES



3.2 EXPERIMENTAL CONFIGURATION

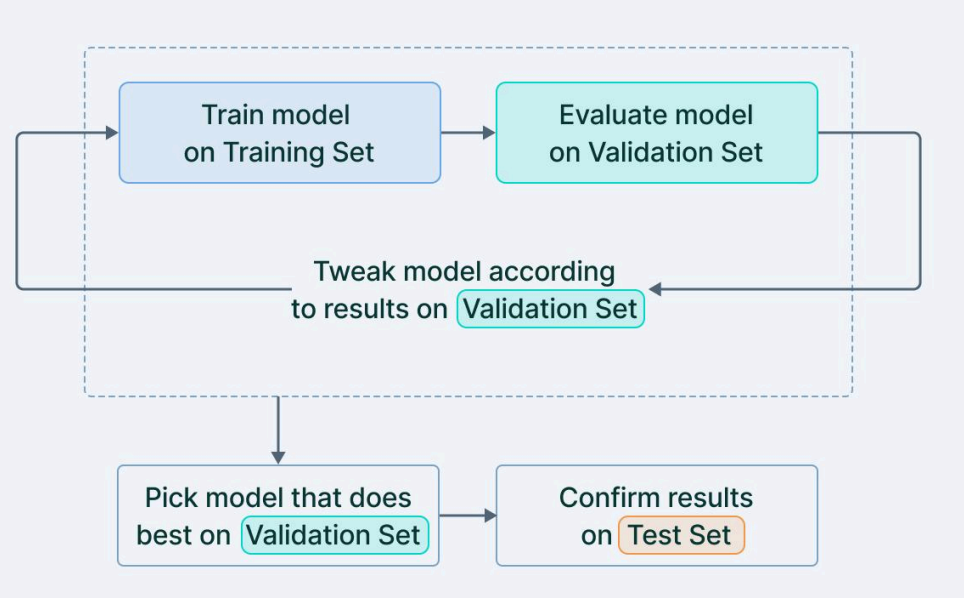


The model was trained using images resized to 128 × 128 pixels, with three colour channels (RGB). The dataset was divided into 80% for training and 20% for validation, using a random seed of 42.

We used a batch size of 32 and trained the model for 10 epochs. Adam was used as the optimizer, with a dropout rate of 0.30 to help reduce overfitting.

The model uses sparse categorical cross-entropy to measure prediction errors and tracks accuracy during training. It predicts three categories: E, O, and R, and has approximately 3.3 million trainable parameters.

3.3 WORKFLOW


Train model
on Training Set
Evaluate model
on Validation Set
Tweak model according
to results on Validation Set
Pick model that does
best on Validation Set
Confirm results
on Test Set

4. DATA PREPROCESSING

4.1 IMAGE RESIZING

Images are resized to 128 × 128 pixels. A consistent input shape is required so that batches can be processed by the same neural-network architecture. Resizing also reduces the computational cost compared with processing large, differently sized source images.

4.2 PIXEL NORMALIZATION

The notebook uses a Rescaling layer with a factor of 1/255 to convert pixel values from the original range of 0–255 to a range of 0–1. This makes the input values smaller and more consistent, helping the neural network process the images more efficiently during training.

4.3 Exploratory Analysis

A grid of sample images is displayed to visually inspect the content and labels. The class counts show substantial imbalance between the R class in TRAIN and the other two classes. The validation subset contains 4 E images, 39 O images and 42 R images. The very small E support makes its class-level metrics unstable: a single incorrect prediction would materially change its recall.
A grid of sample images is displayed to visually inspect the content and labels. The class counts show substantial imbalance be

CNN MODEL DEVELOPMENT

5.1 Using it all over the model, lets actually understand what CNN is?

CNN stands for Convolutional Neural Network and it is a type of deep learning model that analyzes images and automatically learns features such as edges, shapes, colours, and textures.

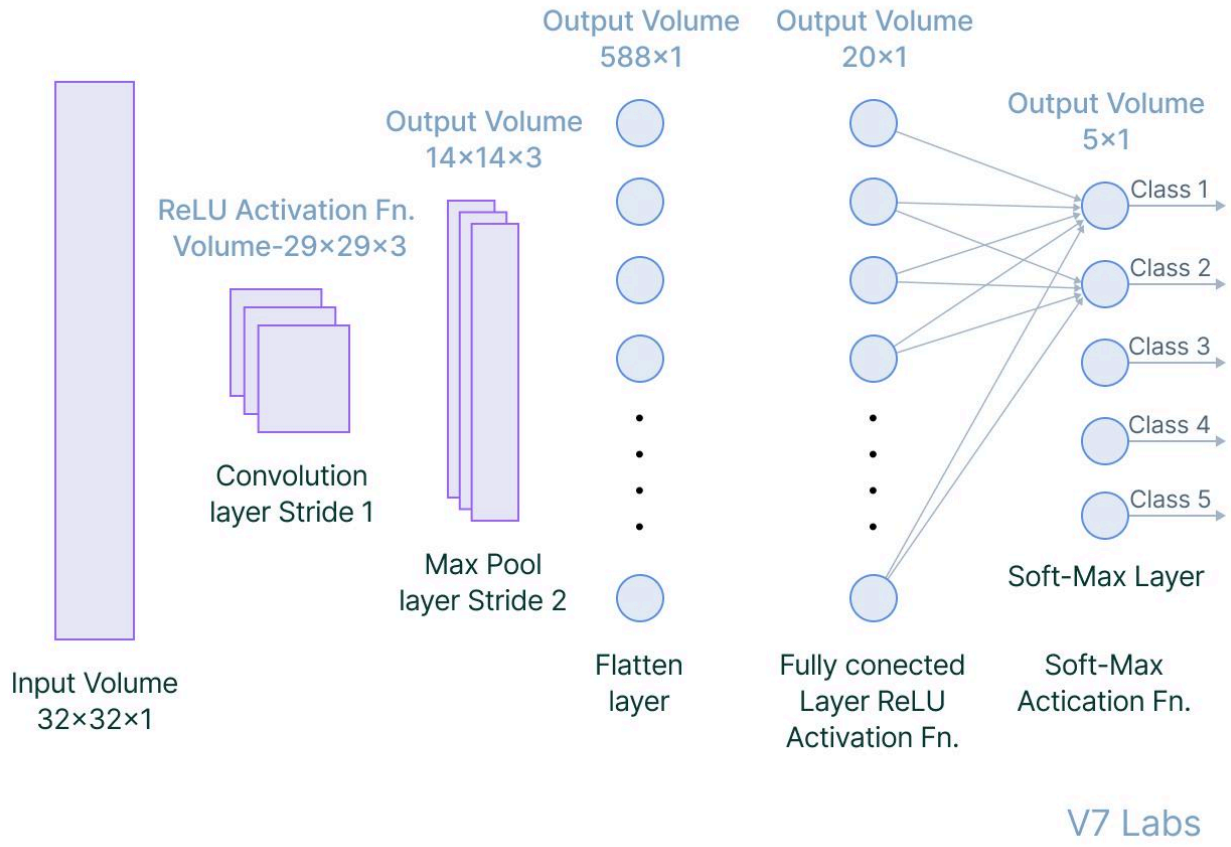
5.2 Why do we require and use CNN here?

We are using CNN in our waste classification project because it can learn visual patterns from waste images without manually defining them, helping classify each image into one of the three categories: E, O, or R. This makes the further process easier.

5.3 TRAINING CONFIGURATION

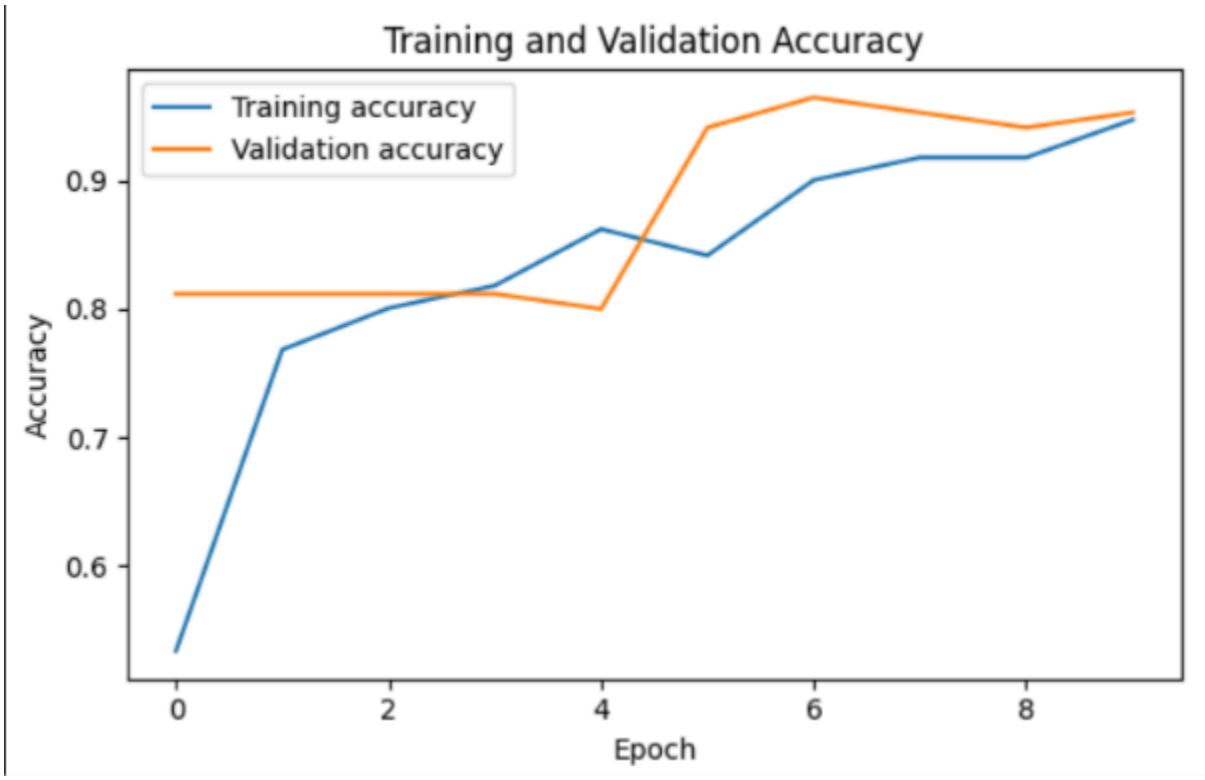
The CNN model is built using TensorFlow/Keras with an input image size of 128 × 128 pixels and a batch size of 32. It is trained for 10 epochs using the Adam optimizer and an 80:20 training-validation split. ReLU and Softmax activation functions, along with sparse categorical cross-entropy loss, help the model learn and classify waste images. Accuracy is used for evaluation, while a dropout rate of 0.3 helps reduce overfitting.

5.4 ARCHITECTURE

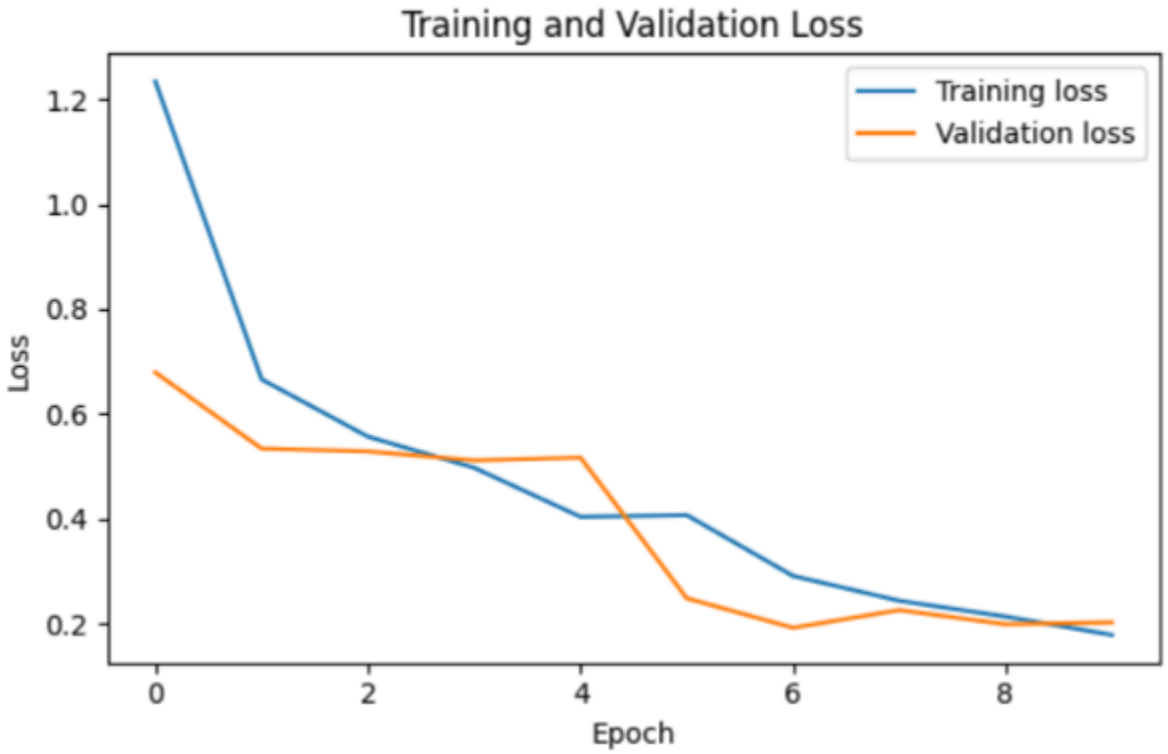


EXPERIMENTAL RESULTS AND EVALUATION

6.1 LEARNING CURVES



The training accuracy improved from 53.37% in the first epoch to 94.72% by the tenth epoch. Validation accuracy started at 81.18%, dropped slightly to 80% in the fifth epoch, and then increased to a peak of 96.47% in the seventh epoch, ending at 95.29%. Although the difference between training and validation accuracy does not automatically mean overfitting or data leakage, the small validation set and an issue with the data split mean that these results should be interpreted carefully.



The training loss steadily decreased from 1.2344 to 0.1777, showing that the model made fewer errors as it learned. The validation loss also improved, reaching its lowest value of 0.1913 in epoch 7 and ending at 0.2016. Although the validation loss stayed close to its best value in the final epochs, testing the model on a separate test dataset is still necessary to check how well it performs on new images.

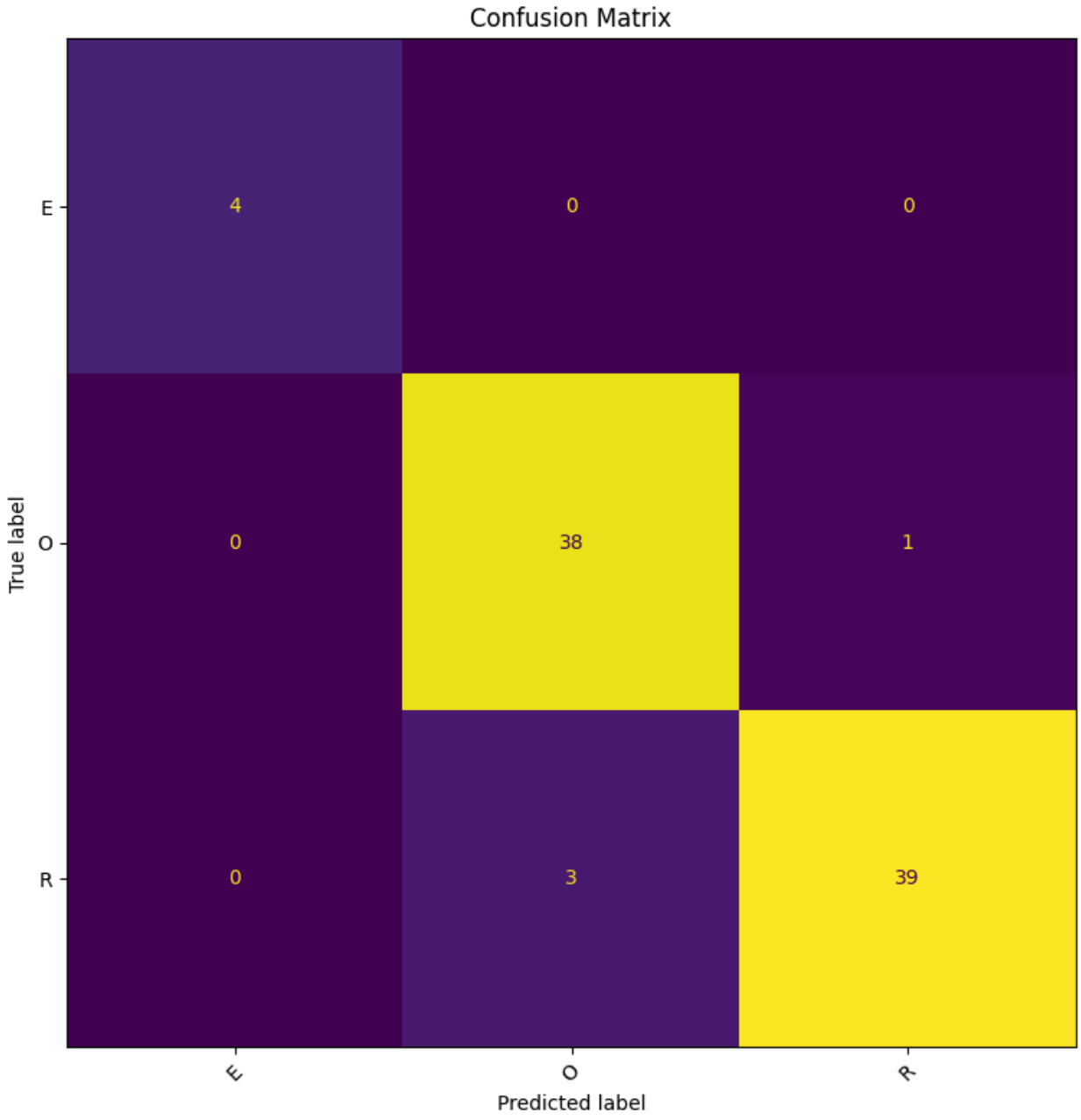
6.2 OVERALL EVALUATION

Metric	Recorded Result	Interpretation
Validation accuracy	95.29%	Correct predictions among 85 validation images.
Validation loss	0.2016	Sparse categorical cross-entropy on validation data.
Macro precision	≈ 0.97	Unweighted average precision across classes.
Macro recall	≈ 0.97	Unweighted average recall across classes.
Macro F1-score	≈ 0.97	Unweighted average of class F1-scores.
Weighted F1-score	≈ 0.95	F1 average weighted by class support.

6.3 CLASS WISE CLASSIFICATION REPORT

Class Label	Precision	Recall	F1-score	Support
E	1.00	1.00	1.00	4
O	0.93	0.97	0.95	39
R	0.97	0.93	0.95	42
Overall accuracy	—	—	0.95	85
Macro average	0.97	0.97	0.97	85
Weighted average	0.95	0.95	0.95	85

6.4 CONFUSION MATRIX



INTERPRETATION

7.1 MAIN FINDINGS

The CNN model achieved 95.29% validation accuracy, showing that it learned useful visual patterns to classify waste into three categories: E, O, and R. The O and R categories achieved similar F1-scores of approximately 0.95. Although E showed perfect reported metrics, only four validation images were available for this category, making the result less reliable.

7.2 EXACT MEANING OF RESULT

The results show that the model performs well on the selected validation data. However, this does not guarantee that it will achieve the same 95% accuracy on images taken in different places or under different lighting and camera conditions. Sometimes, a model may learn patterns in the background or image quality rather than focusing only on the waste material itself. Therefore, testing it on images from different environments is important to understand how well it performs in real-world situations.

7.3 PRACTICAL INTERPRETATION

The application can show the predicted waste category along with its confidence score and a suggested next step. If the prediction is uncertain or the waste contains mixed materials, manual verification should be recommended. This helps avoid incorrect disposal decisions and keeps waste sorting safer.

8.LIMITATIONS AND IMPROVEMENT

8.1 LIMITATIONS

1. Small validation dataset: The model was evaluated on only 85 images, so the results may not fully represent its performance on a larger variety of waste images.

2. Environmental variations: Changes in lighting, camera angles, backgrounds, and image quality may affect the model's predictions.

3. Mixed and contaminated waste: The model may struggle to classify items made of multiple materials or waste that is dirty or damaged.

4. Limited model comparison: The CNN has not been compared with other models, so we cannot conclude that it is the best approach.

5. Need for real-world testing: The model should be tested on a separate, independent dataset to check whether it performs reliably on new images outside the current dataset.

8.2 IMPROVEMENT

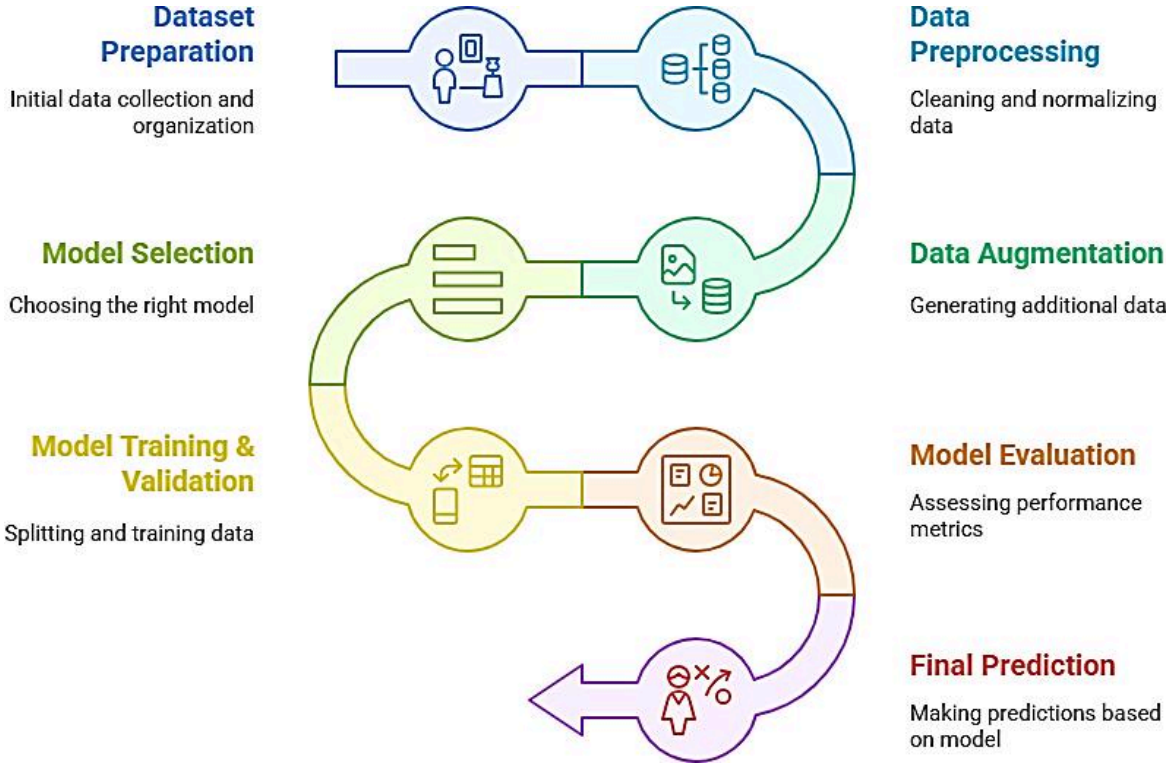
Several steps can make the model more accurate and reliable. First, the data should be improved. The E class has far fewer images than O and R, so collecting more E images, or using class weights or oversampling, would reduce this imbalance. The three duplicate images found between the TRAIN and TEST folders should be removed, and a stratified split should be used so that every class is properly represented in the validation set. Second, data augmentation such as rotation, flipping, zooming and brightness changes can make the model more robust to different lighting, angles and backgrounds. Third, transfer learning with pretrained models such as MobileNetV2 or ResNet50 is likely to give better and more stable results than training a CNN from scratch on a small dataset, since the CNN did not clearly outperform the random forest baseline. Techniques such as early stopping, learning-rate scheduling and k-fold cross-validation can further reduce overfitting and give a more trustworthy estimate of performance than a single 85-image validation set. Finally, the model should be tested on real-world images collected in different environments, and could later be connected to a camera-based automated waste-sorting system, with a confidence threshold that sends uncertain or mixed-material items for manual verification.

9.CONCLUSION

9.1 CONCLUSION

In conclusion, this project shows how CNNs can make waste classification easier and more efficient. The model achieved a validation accuracy of 95.29% and was able to correctly identify most waste categories. This shows that deep learning can be useful in supporting waste segregation and improving recycling practices. However, the model still needs to be tested on a wider variety of images and under different real-world conditions. With further improvements, it could become a helpful tool for smarter waste management and a cleaner environment.

9.2 ML WORKFLOW



REFERENCES

1. LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278–2324. https://doi.org/10.1109/5.726791

2. TensorFlow. (n.d.). Image classification. TensorFlow Tutorials. https://www.tensorflow.org/tutorials/images/classification

3. Project dataset: The waste-image dataset supplied for this project, containing the E, O, and R class folders.

4. Project implementation: Your supplied Jupyter Notebook (`est_miniproject.ipynb`), used for the model architecture, image preprocessing, training, and prediction workflow.

3. Andrew Ng, Machine Learning Specialization, Course 2

4. Python, Jupyter Notebook, pandas, NumPy, Matplotlib and seaborn documentation.